

# Predictive Modeling Plan: Geldium Delinquency Risk

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Task: Predicting Delinquency with AI

## 1. Model Logic & Selection

### Chosen Model: Random Forest Classifier

The Random Forest model was selected because it is an ensemble learning method that handles non-linear relationships and missing data effectively. It provides a great balance between high predictive accuracy and interpretability through feature importance rankings — which is crucial for financial risk assessment.

### Top 5 Input Features

- Credit Utilization Rate** — The most critical indicator of over-extension.
- Payment History (Delayed / Missed)** — Direct evidence of past repayment behavior.
- Debt-to-Income (DTI) Ratio** — Measures the customer's financial breathing room.
- Annual Income** — Determines the fundamental capacity to repay loans.
- Number of Open Credit Lines** — Indicates the level of credit exposure.

## 2. Model Justification

- Excels at handling **noisy data and outliers** identified during EDA (e.g., 100%+ credit utilization).
- Captures **complex interactions** between variables — unlike simple Logistic Regression (e.g., high income combined with high utilization still creates risk).
- Provides **interpretability** through feature importance, supporting regulatory compliance in the financial sector.
- Offers **better stability and scalability** than a single Decision Tree.
- Easier to **deploy and monitor** than a complex Neural Network.

## 3. Model Workflow (Conceptual)

|        |                     |                                                                                          |
|--------|---------------------|------------------------------------------------------------------------------------------|
| Step 1 | Data Ingestion      | Loading the Geldium dataset.                                                             |
| Step 2 | Preprocessing       | Handling missing values via Median / Mean imputation and encoding categorical variables. |
| Step 3 | Feature Engineering | Calculating DTI ratios and scaling numerical inputs.                                     |

|        |            |                                                                                                 |
|--------|------------|-------------------------------------------------------------------------------------------------|
| Step 4 | Training   | Splitting data into <b>80% training</b> and <b>20% testing</b> sets to train the Random Forest. |
| Step 5 | Prediction | Generating a <b>probability score (0 to 1)</b> for delinquency for each customer.               |

## 4. Evaluation Strategy

To ensure the model is both **accurate** and **fair**, the following evaluation metrics will be used:

|                                                 |                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>F1-Score</b>                                 | Since delinquency is often an <b>imbalanced class</b> (fewer defaulters than non-defaulters), F1-Score is better than simple accuracy as it balances Precision and Recall. |
| <b>AUC-ROC</b>                                  | Measures the model's ability to <b>distinguish between at-risk and safe customers</b> across all classification thresholds.                                                |
| <b>Fairness Checks (Disparate Impact Ratio)</b> | Audits the model to ensure it does <b>not unfairly penalize specific demographic groups</b> (based on age or location) if those fields are present.                        |
| <b>Confusion Matrix</b>                         | Specifically monitors <b>False Negatives</b> (at-risk customers missed by the model), as these represent the highest financial loss for Geldium.                           |

The Random Forest Classifier provides the optimal balance of **accuracy, interpretability, and fairness** for Geldium's delinquency prediction task. Combined with rigorous evaluation metrics — especially F1-Score and AUC-ROC — this pipeline will deliver a reliable, production-ready risk scoring system.